

Lab 1 — Introduction to Machine Learning Concepts

Course: ITAI 4376 — AI Applications

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Overview

This lab introduced the foundational concepts of machine learning, including the distinction between supervised, unsupervised, and reinforcement learning. The goal was to build a working understanding of how machine learning models are trained, evaluated, and applied to real-world problems.

What I Learned

Through this lab, I developed an understanding of how data is used to train predictive models. I learned about key concepts such as features, labels, training sets, and test sets. I also explored how different algorithms approach the problem of learning from data, and why choosing the right model matters depending on the type of problem being solved.

Challenges Faced

The primary challenge in this lab was understanding the difference between overfitting and underfitting, and how to recognize each condition when evaluating a model. Interpreting evaluation metrics such as accuracy, precision, and recall also required careful attention, particularly understanding what each metric communicates about model performance beyond simple accuracy.

Tools and Technologies Used

- Python 3
- Jupyter Notebook

- scikit-learn
- pandas
- matplotlib

Key Takeaway

Machine learning is not just about building models — it is about understanding the data, choosing the right approach, and evaluating results in a way that is meaningful for the problem at hand.